

A Function Maps an Input to an Output

A mathematical function is a rule that maps each allowed input to one output.



CORE CONCEPT

Slope Measures Change

The slope of a straight line measures output change per unit of input change. $\text{slope} = \text{change in } y / \text{change in } x = \Delta y / \Delta x$ A positive slope rises, a negative slope falls, and zero slope is horizontal. Slope has units: if x is hours and y is kilometres, slope is kilometres per hour.

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Two points are (1, 3) and (4, 9).

$\Delta y = 9 - 3 = 6$ $\Delta x = 4 - 1 = 3$ $\text{slope} = 6 / 3 = 2$

RUN IT AND INSPECT THE RESULT

Vectors Represent Several Features Together

```
import numpy as np
```

```
a = np.array([2, 4])
```

```
b = np.array([1, 3])
```

```
print(a + b)
```

OUTPUT

```
[3 7]
```

$$2[3, 7] = [6, 14]$$

REASONING SEQUENCE

Matrices Stack Observations into Rows

Frame

one row per
observation;

1

Transform

one column per
feature.

2

X has shape (3, 2): three students, two features. w has shape (2, 1): one weight for each feature. $X @ w$ has shape (3, 1): one result for each student.

CORE CONCEPT

A Dot Product Produces a Weighted Sum

The dot product multiplies matching vector positions and adds the products: $x \cdot w = x_1w_1 + x_2w_2 + \dots + x_nw_n$ In a model, x contains feature values and w contains one weight for each feature. A larger positive weight increases the result when its feature increases; a negative weight decreases it.

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a negative weight decreases it.

$$x = [2, 3] \text{ and } w = [0.4, -0.1]$$

RUN IT AND INSPECT THE RESULT

Variance Measures Squared Distance from the Mean

```
values = np.array([2, 4, 6])  
print(values.var(), values.std())
```

OUTPUT

2.6666666666666665 1.632993161855452

Deviations: [-2, 0, 2] Squared: [4, 0, 4] Variance: $8 / 3 = 2.67$ Standard deviation: $\sqrt{2.67} = 1.63$

CORE CONCEPT

Probability Represents Uncertainty

A probability describes uncertainty about an event. $0 \leq P(\text{event}) \leq 1$ For equally likely outcomes:

0 means impossible under the model.

1 means certain under the model.

0.25 means a 25% probability.

REASONING SEQUENCE

A Derivative Is Local Slope

Frame

A derivative measures how quickly a function changes near one input.

1

Transform

It is the slope of the tangent line at that point.

2

Validate

For $f(x) = x^2$, the derivative is:

3

Explain

At $x = 3$, the local slope is $2 \times 3 = 6$.

4

Exact values: $f(3) = 9$ $f(3.1) = 9.61$ Actual change: 0.61 Derivative estimate: $6 \times 0.1 = 0.6$, which is close for a small step.

RUN IT AND INSPECT THE RESULT

A Gradient Collects One Slope per Parameter

```
gradient = np.array([2.0, -3.0])  
descent_direction = -gradient  
print(descent_direction)
```

OUTPUT

[-2. 3.]

The opposite direction is [-2, 3].

QUALITY GATE

Loss Measures Prediction Error



n : number of examples



y_i : actual target for example i



$\hat{y}_i$: predicted target



$y_i - \hat{y}_i$: residual

QUALITY GATE

Gradient Descent Updates Parameters Step by Step



wold: current parameter



$\partial L / \partial w$: current loss slope



η (eta): learning rate



subtraction moves opposite the gradient

GUIDED LAB

Guided Lab: Match Hand Calculations to NumPy

01

**vector shape and
matrix shape**

02

one dot product

03

**one mean and
population
variance**

04

one probability